

AirNav Risk Analysis Platform (ARAP)

Requirements for an application to conduct safety risk assessments with the 26 methods of the AirNav Risk Analysis Manual

Audience: software developers, architects, testers and the AirNav product owner

Structure follows ISO/IEC/IEEE 29148:2018; quality attributes follow ISO/IEC 25010

Companion to: AirNav Risk Analysis Manual, Edition 0.8

Document control

Item	Detail
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Product owner	Safety Manager, AirNav (to be confirmed)
Source of domain requirements	AirNav Risk Analysis Manual, Ed. 0.8 (the “Manual”). Where this SRS and the Manual differ on method content, the Manual prevails and this SRS shall be corrected.
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Conventions

Requirements use **shall** (mandatory), **should** (recommended) and **may** (optional). Each has a unique identifier, a priority using MoSCoW — **M** must (release 1 blocker), **S** should, **C** could — and a verification method: **T** test, **D** demonstration, **I** inspection, **A** analysis. Identifiers are stable: a deleted requirement keeps its number and is marked “withdrawn”.

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1. Introduction

1.1 Purpose

This document specifies the requirements for the AirNav Risk Analysis Platform (ARAP), a web application that lets AirNav staff plan, conduct, review, approve and maintain safety risk assessments using the 26 methods described in the Manual: Bowtie, HAZID, HAZOP, JHA, FMEA/FMECA, LOPA, FHA, STPA, FTA, bio-mathematical fatigue modelling (three-process model), FRAM and Bayesian Belief Networks (edition 0.1), and collision risk modelling, event tree analysis, human reliability assessment (HEART/CARA), occurrence risk classification (ARMS ERC/EUROCONTROL RAT), GSN safety arguments, common cause analysis, RBD and Markov availability analysis, SWIFT, hierarchical task analysis, simulation evidence, structured expert judgement, security risk assessment and occurrence investigation (SOAM, HFACS, Tripod Beta) (edition 0.2), and species-based wildlife strike risk assessment (edition 0.3). It is written for the development team (in-house or contracted) and for those who will test and accept the product.

1.2 Scope

ARAP replaces spreadsheets and stand-alone drawing tools with one system that holds a single hazard log, applies one risk classification scheme, performs the calculations the methods need, keeps a complete audit trail and generates the Safety Assessment Report. In scope: assessment management, all 26 method modules, calculation engines, hazard log, actions and approval workflow, reporting, libraries, administration and integrations listed in Section 7. Out of scope: occurrence reporting and investigation case management workflow (integrated, not replaced — ARAP holds the risk classification and causal analysis of occurrences), roster production (integrated), and any function used directly for live air traffic control.

1.3 Product goals

- **Consistency:** every assessment follows the process and risk classification scheme of the Manual, whatever method is used.
- **Traceability:** every hazard, control, requirement and action can be traced to its source analysis, its evidence and its approval.
- **Correctness:** quantitative results are computed by verified, versioned engines that reproduce the Manual's worked examples (Section 9).
- **Living risk picture:** results stay connected to safety performance indicators and occurrence data after approval.
- **Usability:** a facilitator can run a live workshop in the tool, projecting worksheets and diagrams, without a separate scribe tool.

1.4 References

Ref.	Document
[M]	AirNav Risk Analysis Manual, Edition 0.8 (draft), September 2026 — method definitions, worksheets, worked examples.
[S1]	ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering.
[S2]	ISO/IEC 25010 Systems and software Quality Requirements and Evaluation (SQuaRE) — product quality model.
[S3]	ISO/IEC 27001:2022 Information security management systems; OWASP Application Security Verification Standard (ASVS).
[S4]	Undang-Undang Nomor 27 Tahun 2022 tentang Pelindungan Data Pribadi (Indonesian Personal Data Protection Law).
[S5]	W3C Web Content Accessibility Guidelines (WCAG) 2.2.
[S6]	ICAO Annex 19, ICAO Doc 9859 (4th ed.), CASR Part 19 (PM 2 Tahun 2026) — regulatory basis of the SMS processes supported.

2. Overall description

2.1 Product perspective

ARAP is a new, self-contained web application with a service-oriented back end. It exchanges data with AirNav's identity provider, occurrence reporting system, rostering system, maintenance management system and document management system (Section 7). Figure 1 shows the logical architecture; it is indicative, and the supplier may propose an equivalent design that meets the requirements.

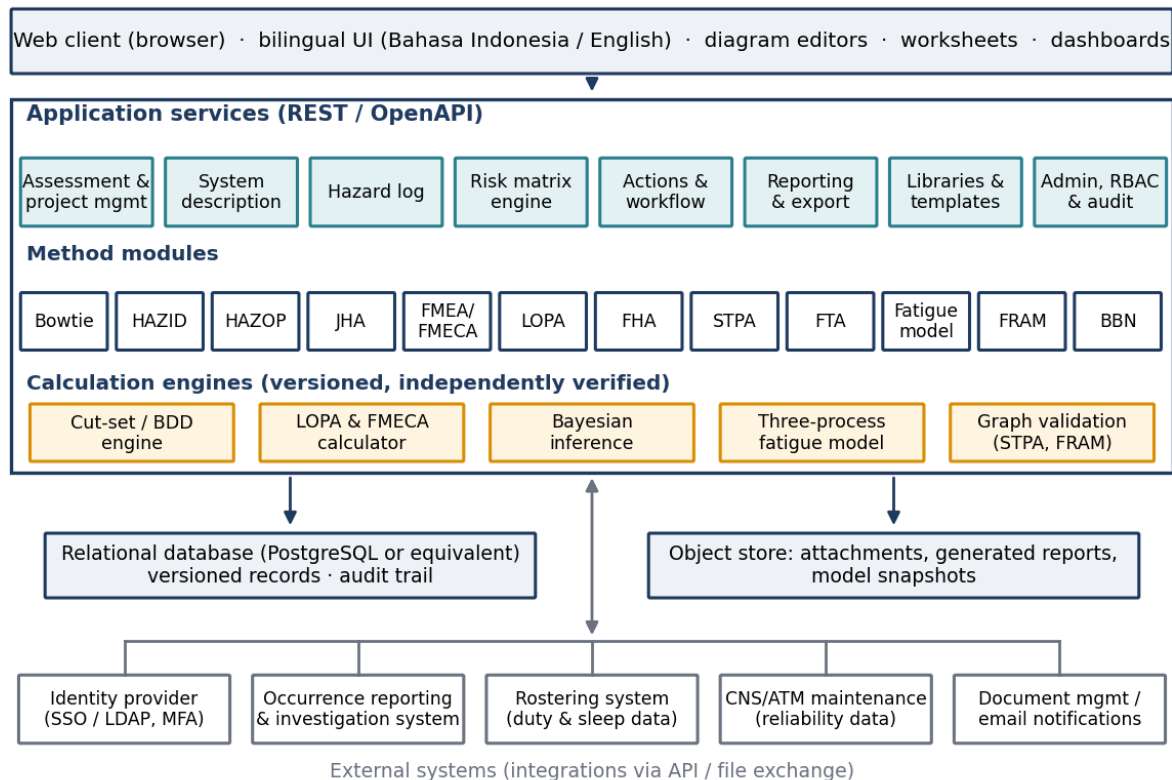


Figure 1. Indicative logical architecture of ARAP.

2.2 User roles

Role	Typical user	Main permissions
Viewer	Any authorised staff member	Read approved assessments, bowties and the hazard log within their organisational scope.
Participant	SME invited to a study	Read the study; comment; contribute to worksheets while the session is open.
Assessor / facilitator	Safety officer, trained facilitator	Create and edit assessments and studies; run sessions; propose risk ratings; submit for review.
Reviewer	Safety Manager or delegate	Review, return with comments, or endorse; edit libraries and templates.
Risk acceptance authority	Unit manager, Director, Accountable Executive	Accept or reject residual risk within their delegated authority (Manual Table 3.3).
Action owner	Any staff member assigned an action	Update status and evidence of their actions.
Fatigue analyst	FRMS specialist	Run fatigue models; access identifiable roster/sleep data (restricted).
Administrator	IT / SQRM admin	Users, roles, organisational structure, risk scheme configuration, integrations. No authority to approve assessments.

2.3 Operating environment

- Browser-based client supporting the two most recent major versions of Chrome, Edge, Firefox and Safari; usable on a 1366 × 768 laptop and projected at 1920 × 1080; read-only views usable on tablets.
- Server deployment on AirNav infrastructure or an approved cloud hosted in Indonesia; containerised deployment (e.g. Docker/Kubernetes) preferred.
- Offline or low-bandwidth use at remote units is supported for data entry by a workshop “offline pack” (Section 4.1, COM-15).

2.4 Constraints and assumptions

- ARAP is a safety-management support tool. It is not used for live ATC and is not subject to ATM software assurance levels; nevertheless its calculation engines shall be verified (Section 6.6) because their results support safety decisions.
- The risk classification scheme, likelihood bands, tolerability regions and acceptance authorities are configuration data, not code (Manual Chapter 3 values are the initial configuration).
- All user interface text shall be available in Bahasa Indonesia and English; stored analysis content is kept in the language in which it was entered.
- Fatigue and roster data are personal data and are processed under [S4].

3. Domain model

The core concept is that every method produces the same kinds of objects — hazards, causes, consequences, controls, risk assessments, requirements and actions — which are stored once and shared. Method-specific objects (a HAZOP node, a fault-tree gate, a BBN node) are linked to these common objects. Figure 2 shows the principal entities.

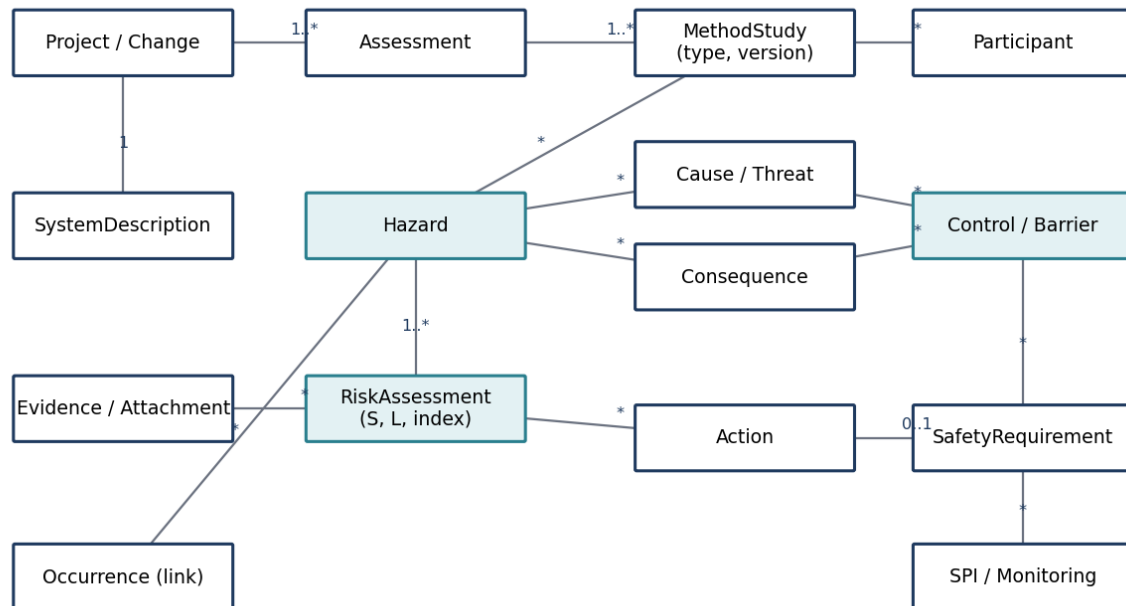


Figure 2. Principal entities and relationships (method-specific entities omitted).

Entity	Key attributes (minimum)
Project / Change	ID, title, type (system, procedure, airspace, organisational, temporary), units affected, sponsor, dates, status
Assessment	ID, project, scope, assumptions, status (draft, in review, endorsed, accepted, closed, superseded), version, approvals
MethodStudy	ID, assessment, method type, method template version, sessions (date, participants, duration), status
Hazard	Global hazard-log ID, title, description, operational context, source studies, owner, status, review date
Cause / Threat; Consequence	ID, text, category (People, Procedures, Equipment, Environment, ...), linked hazard(s)
Control / Barrier	ID, text, type (hardware, software, human, human-hardware, procedure, organisational), side (prevention/recovery), owner, verification status (existing-verified, existing-unverified, planned), effectiveness, criticality, linked SPIs
RiskAssessment	Hazard, consequence, stage (initial, current, residual, target), severity, likelihood, index, region, rationale, evidence links, assessor, timestamp
SafetyRequirement	ID, text, source (FHA objective, LOPA gap, STPA constraint, ...), verification method, status, evidence
Action	ID, text, owner, due date, status, closure evidence, linked hazard/control/requirement
Evidence	File or link, description, source, date, confidentiality marking

4. Functional requirements

4.1 Common functions (COM)

ID	Requirement	Pri.	Verif.
COM-01	The system shall allow an assessor to create a project/change and one or more assessments under it, recording scope, boundary, operational environment, assumptions and before/after description (Manual §2.3).	M	T
COM-02	The system shall allow one or more method studies of any of the 26 method types to be added to an assessment, and shall record which template version each study used.	M	T
COM-03	The system shall recommend candidate methods from the change type and lifecycle phase using the Manual's selection guidance (Manual Table 4.1 and §4.1); the recommendation shall be advisory only.	S	D
COM-04	The system shall provide a session mode for live workshops: participant list and attendance, timer, presentation layout for projection, and capture of dissenting views against any row.	M	D
COM-05	Every hazard, cause, consequence and control created in any method study shall be stored as, or linked to, a common object in the hazard log, with the source study and row recorded.	M	T
COM-06	The system shall detect probable duplicate hazards and controls (text similarity and shared links) and offer to merge them, keeping all source links.	S	T
COM-07	Risk assessments shall use the configured risk classification scheme; the user selects severity and likelihood and the system derives the risk index, tolerability region and required acceptance authority.	M	T
COM-08	The system shall require a rationale and allow evidence links for every severity and likelihood judgement; it shall flag assessments where no evidence is linked.	M	T
COM-09	Only controls with status "existing-verified" shall be credited in the current-risk assessment; planned controls may only be credited in the residual (target) risk.	M	T
COM-10	Where a quantitative method (FTA, LOPA, BBN, FMECA, FHA objectives) produces a frequency or probability, the system shall map it to a likelihood level using the configured quantitative bands and show the unit of measure; comparing values with different units shall be blocked.	M	T
COM-11	The system shall support the workflow Draft → In review → Endorsed → Accepted / Rejected → Closed, with configurable reviewers and acceptance authority derived from the highest residual risk region in the assessment.	M	T
COM-12	Actions shall be assignable to any user with due date and reminders (e-mail and in-app); overdue actions shall be escalated to the assessment owner and the Safety Manager.	M	T
COM-13	Endorsed or accepted assessments shall be locked; changes require a new version that shows a comparison with the previous version.	M	T
COM-14	Comments shall be possible on any object, with mention of users and resolution status.	S	D
COM-15	An offline workshop pack (exported worksheet file) should allow data capture without network access and later re-import with conflict detection.	C	T
COM-16	Global search shall find hazards, controls, requirements, actions and studies by ID, text, unit, system, method and status.	M	T
COM-17	Assessors and reviewers shall be able to archive a project and restore it. An archived project and its assessments shall be hidden from the project and assessment lists unless the user asks to show archived items, shall keep all records readable, and shall not accept new assessments. Archiving and restoring shall be recorded in the audit trail.	M	T
COM-18	Only an administrator shall be able to delete a project, after typing the project code to confirm. Deletion shall remove the project with its assessments, studies, approvals, hazards, controls and actions; it shall be refused while any assessment is endorsed, accepted, closed or superseded (those are safety records and can only be archived); it shall be recorded in the audit trail with a snapshot of the project and the number of records removed; and a deleted demo project shall not be re-created at the next start-up.	M	T

4.2 Hazard log (HZL)

ID	Requirement	Pri.	Verif.
HZL-01	The hazard log shall hold every hazard with the minimum content listed in Manual §18.2 and a unique, never-reused identifier.	M	I
HZL-02	The hazard log shall show, for each hazard, all linked studies, controls, risk assessments (initial, current, residual), requirements, actions, SPIs and occurrences.	M	D
HZL-03	Users shall be able to filter and export the hazard log (XLSX, CSV) by unit, system, risk region, owner, status and review date.	M	T
HZL-04	The system shall notify hazard owners 30 days before a review date and flag overdue reviews on the dashboard.	M	T
HZL-05	Linking an occurrence (from the occurrence reporting system) to a hazard and to the barrier(s) that failed shall be possible, and barrier-failure counts shall feed the bowtie barrier view.	S	T

4.3 Bowtie module (BOW)

ID	Requirement	Pri.	Verif.
BOW-01	The module shall provide a graphical bowtie editor with hazard, top event, threats, consequences, prevention barriers, recovery barriers, escalation factors and escalation-factor barriers (Manual §6.1).	M	D
BOW-02	Barriers shall carry attributes: type, owner, effectiveness (configurable scale, default very good / good / poor / very poor / unknown), criticality, verification status, linked SPIs/audit items and free text.	M	T
BOW-03	The same barrier object may appear on several paths; editing it in one place shall update all occurrences and the diagram shall indicate shared barriers.	M	T
BOW-04	The editor shall automatically lay out the diagram and allow manual adjustment; it shall support collapse/expand of paths and colour display by effectiveness or owner.	M	D
BOW-05	The module shall generate the barrier register table (Manual §6.3) and export the diagram as SVG, PNG and PDF.	M	T
BOW-06	A bowtie library shall hold approved template bowties (e.g. those adapted from the UK CAA templates, subject to their terms of use) that can be copied into an assessment and adapted.	S	D
BOW-07	The module should import bowtie models from a documented JSON schema and from CSV, and should attempt import of BowTieXP XML exports.	S	T
BOW-08	The module shall warn when a consequence path has fewer than two barriers, when a barrier has no owner, or when a top event is phrased as a consequence (keyword check, advisory).	S	T

4.4 HAZID module (HZD)

ID	Requirement	Pri.	Verif.
HZD-01	The module shall provide a HAZID register with columns: ID, guideword/category, hazard, causes, consequences, existing controls, severity, likelihood, risk index, actions, owner (Manual §7.4).	M	T
HZD-02	Guideword checklists shall be configurable libraries; the ANSP checklist of Manual Table 7.2 shall be supplied as default.	M	I
HZD-03	The facilitator shall be able to step through guidewords one at a time in session mode and record “no hazard identified” for a guideword.	M	D
HZD-04	Any row may be flagged “needs detailed method” with the proposed method; the flag shall create a draft study of that type linked to the hazard.	S	T

4.5 HAZOP module (HZIP)

ID	Requirement	Pri.	Verif.
HZIP-01	The module shall allow the design to be divided into nodes, each with a design intent, drawings/attachments and parameters.	M	T
HZIP-02	The module shall generate the deviation matrix from selected parameters × guidewords (default set: No/Not, More, Less, As well as, Part of, Reverse, Other than, Early, Late, Before, After — Manual Table 8.1) and let the team mark each deviation as not meaningful, or analyse it.	M	T
HZIP-03	For each analysed deviation the worksheet shall capture causes, consequences, safeguards, risk index and recommendations, and show node progress (deviations analysed / total).	M	T
HZIP-04	The module shall support procedural HAZOP (steps of a procedure as nodes) and data-flow HAZOP (messages as nodes) with separate default parameter sets.	S	D

4.6 JHA module (JHA)

ID	Requirement	Pri.	Verif.
JHA-01	The module shall record a job (title, location, equipment, frequency, required permits) and its ordered steps, with hazards and controls per step (Manual §9.3).	M	T
JHA-02	Each control shall be classified by the hierarchy of controls (elimination, substitution, engineering, administrative, PPE); the module shall warn when all controls on a step are administrative or PPE.	M	T
JHA-03	A JHA shall be printable as a one-page (A4 landscape) briefing sheet and linkable to work orders in the maintenance system.	S	D
JHA-04	Where a job affects an operational service (e.g. equipment out of service), the module shall prompt for the ATS coordination required and link to the relevant hazard.	S	D

4.7 FMEA / FMECA module (FME)

ID	Requirement	Pri.	Verif.
FME-01	The module shall hold an item hierarchy (system → subsystem → item) importable from CSV/XLSX, with failure rate and data source per item.	M	T
FME-02	The worksheet shall capture, per failure mode: cause, local effect, end effect on ATS, detection method, compensating provisions, severity, occurrence and detection ratings (Manual §10.3).	M	T
FME-03	The module shall compute $RPN = S \times O \times D$ using configurable 1–10 scales that are displayed with their definitions, and shall highlight high-severity modes regardless of RPN.	M	T
FME-04	The module shall compute MIL-STD-1629A failure-mode criticality $C_m = \beta \cdot \alpha \cdot \lambda_p \cdot t$ and item criticality per severity class, with α per item summing to 1 (validation error otherwise).	S	T
FME-05	The module shall list single points of failure and allow export of failure modes as basic events into the FTA module.	S	T

4.8 LOPA module (LOP)

ID	Requirement	Pri.	Verif.
LOP-01	The module shall record scenarios with initiating event and frequency (with unit and data source), conditional modifiers/enabling conditions, candidate safeguards and a tolerable frequency target.	M	T
LOP-02	For each safeguard the user shall answer the four IPL criteria (independent, effective, dependable, auditable) with justification; only safeguards meeting all four may be credited, with a PFD.	M	T
LOP-03	The module shall compute $f_C = f_{IE} \times PP_{CM} \times PPFD_i$, the ratio to target and the additional risk-reduction factor needed, and chart the reduction per layer on a log scale (Manual Figure 5).	M	T
LOP-04	The module shall warn when two credited IPLs share a declared dependency (e.g. same surveillance source, same person) and require an explicit override justification.	S	T
LOP-05	Libraries of initiating-event frequencies and generic PFDs shall be maintainable by reviewers, each value with source and validity date.	S	I

4.9 FHA module (FHA)

ID	Requirement	Pri.	Verif.
FHA-01	The module shall record system functions and, for each, failure conditions of the types total loss, partial loss/degradation, erroneous (detected / undetected), unintended and delayed (Manual §12.1).	M	T
FHA-02	For each failure condition the module shall capture operational effects, context assumptions, severity and the derived safety objective, computed from the configured scheme as the maximum frequency that keeps the risk at or below the configured target region.	M	T
FHA-03	The module shall allow apportionment of an objective among failure conditions contributing to the same outcome, with a check that the apportioned sum does not exceed the objective.	S	T
FHA-04	Safety objectives and derived requirements shall be exportable to the FTA module as top events with targets, and to the requirement register.	M	T

4.10 STPA module (STP)

ID	Requirement	Pri.	Verif.
STP-01	The module shall record losses, system-level hazards (linked to losses) and system-level constraints (linked to hazards) (Manual §13.1 step 1).	M	T
STP-02	A control-structure editor shall allow controllers, controlled processes, control actions and feedback paths to be drawn hierarchically, and each controller to hold a process-model description.	M	D
STP-03	For every control action the module shall generate a UCA table with the four types (not providing; providing; too early/too late/wrong order; stopped too soon/applied too long), allowing “not applicable” with reason.	M	T
STP-04	Each UCA shall require a context (“when ...”) and at least one linked hazard; the module shall report UCAs without context or hazard links.	M	T
STP-05	The module shall record loss scenarios per UCA (types: inadequate feedback, flawed process model, conflicting control, component failure, control action not executed properly) and derive requirements/constraints from them.	M	T
STP-06	A traceability view shall show loss → hazard → constraint → UCA → scenario → requirement, and flag broken chains.	S	D
STP-07	The module should support CAST (accident/incident analysis) reusing the same control structure.	C	D

4.11 FTA module (FTA)

ID	Requirement	Pri.	Verif.
FTA-01	The module shall provide a fault-tree editor supporting top event, intermediate events, basic, undeveloped and house events, and OR, AND, k-out-of-n (voting) and NOT-free logic; transfer gates shall allow a subtree to be reused.	M	D
FTA-02	Basic events shall accept a probability, or a failure rate with exposure time, or an unavailability model (λ , MTTR or test interval), and shall convert them to probability for the chosen mission time, displaying the formula used.	M	T
FTA-03	The engine shall compute minimal cut sets exactly (e.g. MOCUS or binary decision diagrams) for trees up to 2 000 basic events, with an order/probability truncation option for larger trees, and report truncation.	M	T
FTA-04	The engine shall compute the top-event probability both exactly (BDD) and by the rare-event approximation, and display both when they differ by more than 1%.	M	T
FTA-05	The engine shall compute Birnbaum and Fussell-Vesely importance for every basic event and rank cut sets by contribution.	M	T
FTA-06	Common-cause failure groups shall be definable with the β -factor model, and the engine shall add the CCF events to the cut sets automatically.	S	T
FTA-07	The module shall compare the result with a linked FHA objective and show pass/fail and margin; sensitivity analysis shall vary any input over a user-defined range.	S	T
FTA-08	Fault trees shall be exportable as SVG/PDF and in the Open-PSA Model Exchange Format.	S	T

4.12 Fatigue modelling module (FAT)

ID	Requirement	Pri.	Verif.
FAT-01	The module shall import duty schedules from the rostering system (API or CSV) and allow manual entry of rosters and roster templates.	M	T
FAT-02	Sleep periods shall be generated by configurable rules (e.g. normal night sleep window, day sleep after night duty, optional naps) or imported from sleep diaries/actigraphy; generated and measured sleep shall be distinguished.	M	T
FAT-03	The engine shall implement the three-process model as specified in Section 5.1, with all parameters configurable and the default values of the Manual (Table in §15.2).	M	T
FAT-04	For each duty the module shall report minimum alertness and its clock time, maximum KSS, time and share of duty at or above a configurable KSS threshold (default 7), and plot alertness over time.	M	T
FAT-05	Users shall be able to compare two or more roster variants side by side and export the comparison.	M	D
FAT-06	Aggregated, anonymised roster-level results shall be the default view; identifiable individual results shall be visible only to the Fatigue analyst role and logged on access.	M	T
FAT-07	The module should allow calibration of parameters against collected KSS data (e.g. fitting the KSS intercept and slope) and store calibrated parameter sets with version and date.	C	T
FAT-08	Outputs shall carry the notice that model predictions are group averages and not a measure of an individual's fatigue.	M	I

4.13 FRAM module (FRM)

ID	Requirement	Pri.	Verif.
FRM-01	The module shall provide a FRAM editor with hexagonal functions and the six aspects (Input, Output, Precondition, Resource, Control, Time), each aspect holding free-text entries.	M	D
FRM-02	Couplings shall be created by matching an Output text to an aspect of another function; the module shall list unmatched aspects ("orphans") for validation.	M	T
FRM-03	Each function shall have a type (human, technological, organisational) and, per instantiation, output variability for timing (too early, on time, too late, not at all) and precision (precise, acceptable, imprecise).	M	T
FRM-04	The module shall support instantiations (scenarios) that highlight the active couplings and propagate variability downstream for review.	S	D
FRM-05	Models should be importable and exportable in the FRAM Model Visualiser file format.	S	T

4.14 BBN module (BBN)

ID	Requirement	Pri.	Verif.
BBN-01	The module shall provide a graph editor for discrete-state nodes and directed arcs and reject cycles.	M	T
BBN-02	A CPT editor shall enforce that each column sums to 1 (within 1e-9), support copy/paste from spreadsheets, and record the source of each table (data, expert elicitation with names/roles, literature).	M	T
BBN-03	The engine shall perform exact inference (variable elimination or junction tree) for networks up to 60 nodes with up to 5 states and 4 parents per node, returning marginals within 2 s.	M	T
BBN-04	Users shall be able to set hard evidence on any node and see updated marginals (predictive and diagnostic queries), and save named scenarios.	M	T
BBN-05	The module shall provide sensitivity analysis (tornado chart) of a target node to each CPT parameter or parent node.	S	T
BBN-06	Parametric CPT helpers (noisy-OR, ranked nodes) should be provided.	C	T
BBN-07	Networks shall be exportable/importable in XMLBIF, and should be importable from GeNIe (.xdsl).	S	T
BBN-08	Bowtie threats and barriers, and FTA structures, may be converted into a BBN skeleton for further editing.	C	D

4.15 Collision risk module (CRM)

ID	Requirement	Pri.	Verif.
CRM-01	The module shall compute collision risk with the Reich model in the occupancy form of ICAO Doc 9689 (§5.5) for the vertical and lateral dimensions, reporting same- and opposite-direction components and the total.	M	T
CRM-02	Aircraft dimensions shall be entered in feet or metres and converted to NM internally; every parameter shall carry its source and date.	M	I
CRM-03	The module shall compare the result with a configurable TLS (defaults: 2.5×10^{-9} RVSM technical, 5×10^{-9} total vertical and lateral) and show the ratio to the TLS.	M	T
CRM-04	For lateral risk the module shall derive $P_y(S_y)$ from a Gaussian or double-exponential deviation model, plot risk against route spacing on a log scale and solve for the minimum spacing meeting the TLS.	M	T
CRM-05	Sensitivity runs (traffic growth, deviation scale, tail model) should be saved as named cases.	S	D

4.16 Event tree module (ETA)

ID	Requirement	Pri.	Verif.
ETA-01	The module shall accept an initiating event with frequency and unit and an ordered list of barriers with success probabilities, and draw the event tree.	M	D
ETA-02	Conditional success probabilities shall be definable per path prefix (dependence); the tree shall support barrier trees (success ends the sequence) and full functional trees.	M	T
ETA-03	The engine shall compute probability and frequency per sequence, check that end-state probabilities sum to 1 ($\pm 1e-9$), and aggregate frequency by severity.	M	T
ETA-04	Each end state shall carry an outcome label and severity, and may be sent to the hazard log.	S	T
ETA-05	An ETA should be linkable to the bowtie whose consequence side it quantifies, and use FTA or HRA results as inputs.	C	D

4.17 Human reliability module (HRA)

ID	Requirement	Pri.	Verif.
HRA-01	The module shall provide the HEART and CARA libraries (generic task types with nominal HEP; error-producing conditions with maximum effect) as versioned reference data.	M	I
HRA-02	For each task the user shall choose the library and GTT, add EPCs with an APoA in [0,1] and a justification; the HEP shall be computed by §5.6 and capped at 1.	M	T
HRA-03	The module shall show each EPC's factor and rank the factors to support improvement.	M	D
HRA-04	HEPs shall be referenceable from FTA basic events, ETA branches, LOPA IPLs and HTA tasks.	S	D

4.18 Occurrence risk classification module (ORC)

ID	Requirement	Pri.	Verif.
ORC-01	The module shall classify occurrences with the ARMS ERC matrix by answering Q1 (most credible accident outcome) and Q2 (effectiveness of remaining barriers) and display the risk index and band.	M	T
ORC-02	Band thresholds shall be configurable (defaults ≥ 100 red, ≥ 20 amber).	M	T
ORC-03	The module shall provide the EUROCONTROL RAT marksheet for risk of collision and controllability, computing the score; the ESARR 2 severity class shall be recorded by the analyst.	M	T
ORC-04	Red-band occurrences and severity classes A and B shall be flagged for investigation and may be sent to the hazard log.	M	T
ORC-05	Occurrences shall be importable from the occurrence reporting system (Section 7).	S	T

4.19 Safety argument module (GSN)

ID	Requirement	Pri.	Verif.
GSN-01	The module shall provide a GSN editor with goal, strategy, solution, context, assumption and justification elements, SupportedBy and InContextOf links, and the undeveloped marker, laid out automatically.	M	D
GSN-02	Solutions shall link to ARAP studies, hazard log entries or external documents, and show the status of the linked evidence.	M	T
GSN-03	The module shall check structure: one top goal; no cycles; strategies supported only by goals; solutions and contextual elements as leaves; undeveloped or unsupported goals and solutions without evidence reported.	M	T
GSN-04	The argument should be exportable as SVG and included in the Safety Assessment Report.	S	T

4.20 Common cause analysis module (CCA)

ID	Requirement	Pri.	Verif.
CCA-01	The module shall provide ZSA, PRA and CMA worksheets with the columns of the Manual (§23.3), including a configurable particular-risk list and common-mode source list.	M	I
CCA-02	Independence claims shall be importable from the AND and k-out-of-n gates of an FTA study.	M	T
CCA-03	Claims judged not independent, and particular risks that defeat redundancy, shall be sendable to the hazard log.	M	T

4.21 Availability module (RBD)

ID	Requirement	Pri.	Verif.
RBD-01	The module shall provide an RBD editor with nested series, parallel and k-out-of-n groups and blocks with MTBF and MTTR.	M	D
RBD-02	The engine shall compute availability, unavailability, downtime per year and, per block, the availability gain if the block were perfect (§5.7).	M	T
RBD-03	The module shall solve continuous-time Markov models (states marked up/down, transition rates) for steady-state probabilities, availability, failure frequency, mean down time and mean time to first failure.	M	T

4.22 SWIFT module (SWF)

ID	Requirement	Pri.	Verif.
SWF-01	The module shall provide a SWIFT worksheet with configurable prompt categories and risk classification per row.	M	I
SWF-02	The module shall show category coverage and allow “nothing credible identified” to be recorded per category.	M	T
SWF-03	Rows with a consequence shall be sendable to the hazard log.	M	T

4.23 Task analysis module (HTA)

ID	Requirement	Pri.	Verif.
HTA-01	The module shall maintain a task hierarchy with automatic numbering, plans, and tree and outline views.	M	D
HTA-02	Tasks with sub-tasks but no plan shall be flagged.	M	T
HTA-03	Bottom-level tasks shall carry error modes from a configurable taxonomy and may link to a task in an HRA study, displaying its HEP.	M	T

4.24 Simulation evidence module (SIM)

ID	Requirement	Pri.	Verif.
SIM-01	The module shall record the exercise plan (type, objectives, scenarios, participants, limitations).	M	I
SIM-02	For each measure the user shall define the better direction and a success criterion (threshold or no worse than baseline plus margin) before entering data.	M	T
SIM-03	The engine shall compute n, mean, standard deviation and 95% confidence interval per condition, Welch's t-test for the difference, and whether the criterion is met (§5.8).	M	T
SIM-04	Measures that fail their criterion shall be sendable to the hazard log.	S	T

4.25 Expert judgement module (SEJ)

ID	Requirement	Pri.	Verif.
SEJ-01	The module shall record experts, seed and target questions (with scale linear/log) and each expert's 5/50/95% quantiles, validating that quantiles increase.	M	T
SEJ-02	The engine shall implement Cooke's classical model (§5.9): calibration, information, weights with significance level α , and performance-weighted and equal-weight decision makers.	M	T
SEJ-03	The module shall support Delphi rounds and report median, quartiles, IQR and convergence per round.	M	T
SEJ-04	All elicitation inputs and results shall be retained and reproducible.	M	I

4.26 Security risk module (SRA)

ID	Requirement	Pri.	Verif.
SRA-01	The module shall provide a register of threat scenarios (asset, threat, threat source, vulnerability, C/I/A impact 1-5, likelihood 1-5, controls, treatment, owner).	M	I
SRA-02	The security risk shall be computed as $L \times \max(C, I, A)$ with configurable levels (defaults ≥ 15 very high, ≥ 10 high, ≥ 5 medium).	M	T
SRA-03	Each scenario shall carry a safety severity; only scenarios with a safety effect shall be sent to the safety hazard log, without exploit detail.	M	T
SRA-04	Access to security studies shall be restrictable to named users (need to know).	S	T

4.27 Investigation module (INV)

ID	Requirement	Pri.	Verif.
INV-01	The module shall record occurrence facts and build a SOAM analysis: barriers (six types; failed, absent or effective), human involvement, contextual conditions (five categories), organisational factors (twelve categories) and actions, and draw the SOAM chart.	M	D
INV-02	Failed and absent barriers shall be linkable to bowtie barriers; each bowtie barrier shall show how many investigations recorded it as failed.	M	T
INV-03	The module shall provide the HFACS taxonomy (four tiers, 19 categories) with evidence per selected category.	M	T
INV-04	The module shall provide Tripod Beta event trios with barrier failure chains (immediate cause, precondition, underlying cause) and Basic Risk Factor classification, with a BRF profile.	M	T
INV-05	Investigation findings shall be sendable to the hazard log.	S	T

4.28 Wildlife strike risk module (WLD)

ID	Requirement	Pri.	Verif.
WLD-01	The module shall record, per species or species group, common and scientific name, body mass, flocking behaviour, and strikes and damaging strikes per year for a configurable review period, with aircraft movements per year.	M	T
WLD-02	The engine shall compute per species the mean strikes per year, the damaging percentage, the strike rate per 10,000 movements and its trend, the likelihood and severity classes, the risk score and level, and a ranking (§5.11).	M	T
WLD-03	Where a species has fewer strikes than a configurable minimum (default 5), severity shall be derived from body mass with one class added for flocking species, and the basis shall be shown.	M	T
WLD-04	Class bands and risk thresholds shall be configurable per aerodrome and recorded with the result.	S	T
WLD-05	The module shall show the species on a 5 × 5 likelihood-severity matrix and hold a profile per species: ecology notes, monthly survey counts and strikes, time of day, and species management measures with type, owner and status.	M	D
WLD-06	The module shall keep a register of wildlife attractants with land use, distance and species attracted, flagging those within the configurable assessment radius (default 13 km, ICAO Annex 14 §9.4).	M	T
WLD-07	High and moderate species shall be sendable to the hazard log with their attractants and measures.	M	T
WLD-08	Strike records should be importable from the occurrence reporting system and in ICAO IBIS format.	C	T

4.29 Assessment templates (TPL)

ID	Requirement	Pri.	Verif.
TPL-01	The system shall offer a catalogue of assessment templates; creating an assessment from a template shall pre-fill scope, environment and assumptions and create the template's method studies with proposed content.	M	T
TPL-02	Template content shall be proposed, not rated: likelihoods, calculations and hazard log entries shall be left for the assessment team, and the user shall be told this when choosing the template.	M	I
TPL-03	Cross-links inside a template (GSN solutions to studies, HTA tasks to HRA tasks) shall point to the studies created for the new assessment.	M	T
TPL-04	The system shall provide the template "New AIM system acquisition" of Manual Appendix C: FHA (8 functions, 12 failure conditions), data-flow HAZOP (7 nodes incl. migration), FMEA, FTA, CCA, STPA, HTA, HRA, security, JHA, SWIFT, transition bowtie, shadow-operation measures and GSN skeleton.	M	T
TPL-05	Administrators should be able to add templates and save an existing assessment as a new template, with version control.	C	D
TPL-06	The system shall provide the template "Major organisational change" of Manual Appendix D: function map (18 functions incl. SMS roles and barrier owners, checks unrated), FHA, STPA of the current and new structures, HAZID with its own organisational guideword list, HTA of a merged post, Delphi, SWIFT, transition bowtie, SPI register and GSN argument.	M	T
TPL-07	A HAZID or SWIFT study shall be able to carry its own guideword list, used for its guideword options and coverage in place of the method default.	M	T

4.30 Organisational function map and SPI register (OFM, SPI)

ID	Requirement	Pri.	Verif.
OFM-01	The function map shall record, per safety-related function, category, current owner, new owner and the checks competence, capacity, authority and handover with the values of Manual Appendix D.	M	T
OFM-02	The system shall flag each row with its gaps — no new owner, competence gap, capacity stretched or overloaded, authority pending or none, handover not planned, any value still unknown — and summarise gaps, orphans and rows still to assess.	M	T
OFM-03	Rows with gaps shall be sendable to the hazard log; rows only "to assess" shall not.	M	T
ID	Requirement	Pri.	Verif.
SPI-01	The SPI register shall record indicator, type (leading / lagging), linked barrier or hazard, baseline, target, alert level, frequency, owner, current value, status (green / amber / red) and escalation.	M	T

ID	Requirement	Pri.	Verif.
SPI-02	Indicators at red shall be listed with their escalation; the register shall be exportable and included in the report.	M	T
SPI-03	SPIs should be linkable to bowtie barriers so that barrier health shows the SPI status.	S	D

4.31 Reporting and dashboards (REP)

ID	Requirement	Pri.	Verif.
REP-01	The system shall generate a Safety Assessment Report (DOCX and PDF) following the structure of Manual §32.3, including selected worksheets, diagrams and calculations, from an administrable template.	M	T
REP-02	Every generated report shall state the assessment version, the calculation-engine versions and the risk-scheme version used.	M	I
REP-03	A dashboard shall show: hazards by residual risk region and unit; overdue actions and reviews; barrier health (effectiveness and linked SPI status) for the bowtie library; assessments by status.	M	D
REP-04	All tables shall be exportable to XLSX/CSV and all diagrams to SVG/PNG.	M	T

4.32 Administration (ADM)

ID	Requirement	Pri.	Verif.
ADM-01	Administrators shall configure the risk classification scheme: severity and likelihood levels, labels (ID/EN), quantitative bands, matrix regions, colours and acceptance authorities; changes shall be versioned and never alter locked assessments.	M	T
ADM-02	Administrators shall maintain the organisational structure (directorates, branches, units) used for scoping permissions and reports.	M	T
ADM-03	Method templates (guideword lists, worksheet columns, rating scales) shall be configurable and versioned by Reviewers.	S	T

4.33 Investigation expert system (IES)

Implements Manual Appendix E. Requirements IES-01 to IES-09 cover the case library and inference (phase 1) and the Bayesian network (phase 2); IES-10 to IES-14 cover drafting AcciMaps from reports (phase 3) and data protection.

ID	Requirement	Pri.	Verif.
IES-01	The system shall hold a case library per occurrence category: each case an AcciMap of evidence, hypothesis and finding blocks in the five ORLIO layers, with ORLIO factor codes from the category's factor catalogue, the state of each block in the past occurrence, input and context links, source, summary and report text.	M	T
IES-02	An AcciMap editor shall allow blocks to be added, edited, linked (input or context) and moved between layers; links that form a cycle shall be rejected.	M	T
IES-03	The system shall generate a default rule set for every hypothesis and finding block (confirming, weakening and contradicting rules, Manual Appendix E.4) and allow the rules to be edited; rules out of date with the links shall be flagged.	M	T
IES-04	Search shall rank approved cases of a category against a free-text narrative using a thesaurus and the factor catalogue (§5.14), and show the matched terms and factors for each case.	M	T
IES-05	The investigator shall rate evidence on the five-term scale or leave it not provided; a rating given to a factor shall apply to every case that uses the factor.	M	T
IES-06	The system shall run fuzzy inference (§5.12) on each selected case and show, per block, the inferred term and value or "not determined".	M	T
IES-07	The system shall list as clues the not-provided evidence needed by undetermined blocks, with the cases that need it and a priority.	M	T
IES-08	The system shall learn a Bayesian network per category from the approved cases, compute the posterior over mechanisms from the ratings and rank unrated factors by value of information (§5.13).	M	T

ID	Requirement	Pri.	Verif.
IES-09	The learned network, with the ratings as evidence, shall be exportable as a Bayesian-network study giving the same posteriors.	S	T
IES-10	The system shall draft an AcciMap from report text or a text-based PDF with an offline drafter (factor catalogue and mechanism keywords).	M	T
IES-11	Where an administrator has configured it, the system should draft AcciMaps with a large language model returning a fixed structure; the model, date and engine shall be recorded with the case.	S	T
IES-12	Every drafted block shall carry a quote from the report; quotes not found in the report text shall be flagged, and a case with a flagged quote shall not be approvable.	M	T
IES-13	A case shall be used by search, inference and the network only after approval by a reviewer; editing an approved case shall return it to draft. Approval shall require at least one evidence block, one finding rating a mechanism Probable or higher, and rules for every derived block.	M	T
IES-14	The user interface shall state that the system is advisory and that its outputs do not apportion blame; the AI drafter shall warn that only published reports may be sent to an external service; case changes, approvals and drafts shall be audited.	M	I
IES-15	Factor catalogues shall carry the ATSB safety factor type code; AcciMaps shall have a technical-failure-mechanism lane; evidence may be rated “unsure” (no evidence, not a clue); verbal probabilities shall use the ATSB terms.	M	T

4.34 Safety investigation analysis — ATSB method (ATB)

Implements Manual Appendix F, after the ATSB Safety Investigation Guidelines Manual — Analysis (v1.07) and Tools Manual — Analysis Supplement (v1.05).

ID	Requirement	Pri.	Verif.
ATB-01	The system shall provide a sequence of events list with start time (mandatory), end time, title, comments, source, theme and display flag, a timeline chart by theme, and promotion of an event to the safety factors list as an occurrence event, individual action, local condition or positive action.	M	T
ATB-02	The system shall keep a safety factors list with title, type, ATSB codes, further-analysis decision (a reason required when “no”), safety-issue flag, existence result, derived finding type, risk level and analysis-complete flag.	M	T
ATB-03	Safety factor types shall be occurrence event, individual action, technical failure mechanism, local condition, risk control, organisational influence, positive action and positive condition; codes shall come from the ATSB aviation taxonomy (Tools App. D), with role and error type for individual actions, functional area for conditions, preventive/recovery for risk controls and internal/external for organisational influences.	M	T
ATB-04	For each factor the system shall record tests for existence and influence (with the factor or occurrence influenced) as evidence tables — items rated supports, opposes, no effect or unsure, with comments, source, evidence type, relevance, credibility concerns and expectation — each with a conclusion, a probability expression and a summary; and a test for importance with a justification.	M	T
ATB-05	The system shall derive the finding type (§5.15): contributing safety factor, other safety factor, positive safety factor, not established, not a safety factor, pending or excluded; findings shall require the standard of proof “likely” ($\geq 66\%$).	M	T
ATB-06	Only local conditions, risk controls and organisational influences shall be flaggable as safety issues; a safety issue shall carry an owner and a risk analysis: worst possible scenario, existing controls and effectiveness, worst credible scenario, consequence and likelihood with justifications, and a sensitivity rating.	M	T
ATB-07	Safety issues shall be rated on the AirNav risk scheme by default, or on the ATSB 6×6 as an option chosen per safety issue; both shall yield critical, significant or broadly acceptable (not a safety issue).	M	T
ATB-08	The system shall record safety actions per safety issue (kind, organisation, notification date, communication log, status, ATSB safety action classification), the residual risk, the ALARP judgement and the practicability considerations, and derive whether further action is required; the safety issue status shall be pending, adequately, partially or not addressed.	M	T
ATB-09	While a safety issue is above ALARP and not adequately addressed, the system shall compute a follow-up date six months after the last contact and flag it when overdue.	S	T

ID	Requirement	Pri.	Verif.
ATB-10	The system shall check the analysis and list: judgemental or vague wording and bundled factors; arguments from ignorance; conclusions below the standard of proof; compound existence × influence below 50 %; non-independent evidence; invalid influence targets; classification gaps; exclusions without a reason; the test for sufficiency; and fairness for individual actions without identified reasons.	M	T
ATB-11	The system shall provide basic evidence tables for other key findings (other key finding, intermediate finding, positive safety factor, possible scenario).	M	T
ATB-12	The system shall generate a safety factor map in the ORLIO lanes from the verified influences.	S	D
ATB-13	The system shall provide the analysis review checklist (completeness, fairness, organisation of findings, bias), record the stop rule, and organise findings as contributing safety factors, other safety factors and other key findings.	M	T
ATB-14	Safety issues shall be sendable to the hazard log and safety actions to the action list.	M	T
ATB-15	The checklists of Tools App. C (events to look for, questions per level, test questions, evidence criteria) shall be shown where the investigator works.	S	I
ATB-16	The system shall generate an investigation report from an ATSB-method study with the sections: 1 Executive summary (synopsis what/when/where/who, consequences, core finding); 2 Factual information (occurrence timeline from the sequence of events, personnel and assets, immediate actions); 3 ORLIO analysis framework (3.1 occurrence events incl. technical failure mechanisms, 3.2 individual actions, 3.3 local conditions, 3.4 risk controls, 3.5 organisational influences, each with its definition, narrative and factors with evidence); 4 Findings and contributing factors in priority order with the chain to the occurrence; 5 Safety recommendations and corrective actions; 6 Appendices (A photographs and diagrams, B interview summaries, C ORLIO factor map).	M	T
ATB-17	Section 5 shall be a table with the columns ORLIO layer, identified deficiency, corrective action required (prefixed with its hierarchy-of-controls level: elimination, substitution, engineering, administrative, PPE), action owner and target date ([Date] when not set); every contributing factor without a corrective action shall appear as a flagged row and as a note in the checks.	M	T
ATB-18	The system shall record for individual and positive actions why the action made sense to the person at the time, and corrective actions (hierarchy level, target date, reference) for any analysed factor, not only safety issues.	M	T
ATB-19	The report shall be previewable in the editor from unsaved content and downloadable as DOCX and PDF; downloads shall be audited; uploaded images (≤ 3 MB each) shall be stored with the study.	M	T

5. Calculation specifications

The engines below shall be implemented as separately versioned components with documented APIs, so they can be tested independently of the user interface.

5.1 Three-process model of alertness

Time t is in hours, simulated on a fixed step Δt (default 5 min). At each step the state is sleep or wake. Let S be homeostatic level:

Quantity	Update / formula	Default parameter
S while awake	$S \leftarrow L_a + (S - L_a) \cdot e^{d \cdot \Delta t}$	$L_a = 2.4, d = -0.0353$
S while asleep	$S \leftarrow H_a - (H_a - S) \cdot e^{g \cdot \Delta t}$; optional brake: above $S = 12.2$ recovery follows the braked form of Åkerstedt & Folkard	$H_a = 14.3, g = -0.381$, brake off by default
C	$C(t) = C_a \cdot \cos(\pi(t - p)/12)$, t as clock time	$C_a = 2.5, p = 16.8$
W (awake only)	$W = W_c \cdot e^{W_d \cdot \tau}$, τ = hours since last wake	$W_c = -5.72, W_d = -1.51$
U (optional)	12-hour ultradian component	Off by default
Alertness	$A = S + C + W (+ U)$, reported only during wake	—
KSS	$KSS = a + b \cdot A$	$a = 10.6, b = -0.6$
Initial condition	S at simulation start (user-set or spin-up of ≥ 3 days)	$S_0 = 13.0$

5.2 Fault tree quantification

- Minimal cut sets by MOCUS or BDD; results must be identical for trees without truncation.
- Top-event probability: exact via BDD; rare-event approximation $\sum P(\text{cut set})$; min-cut upper bound $1 - \prod(1 - P(\text{cut set}))$.
- Rate conversion: $P = 1 - e^{-\lambda t}$; unavailability of a repairable component $Q \approx \lambda \cdot \text{MTTR} / (1 + \lambda \cdot \text{MTTR})$; periodically tested standby component $Q \approx \lambda \cdot T / 2$.
- Birnbaum $I_B(e) = P(\text{top} | e = 1) - P(\text{top} | e = 0)$; Fussell-Vesely $I_{FV}(e) = [P(\text{top}) - P(\text{top} | e = 0)] / P(\text{top})$.
- β -factor CCF: for a group with rate λ , independent part $(1 - \beta)\lambda$ per member and one common event $\beta\lambda$.

5.3 LOPA, FMECA and risk mapping

- LOPA: $f_C = f_{IE} \times \text{PP}_{CM} \times \text{PPFD}_i$; required additional risk reduction = $\max(1, f_C / f_{\text{target}})$.
- FMEA RPN = $S \times O \times D$; FMECA $C_m = \beta \cdot \alpha \cdot \lambda_p \cdot t$; item criticality $C_r = \sum C_m$ for modes in a severity class.
- Frequency-to-likelihood mapping uses the configured bands with inclusive lower and exclusive upper bounds, after checking unit compatibility.

5.4 Bayesian inference

Exact inference by variable elimination or junction tree; results must agree with brute-force enumeration of the joint distribution to within $1e-9$ for networks of up to 12 binary nodes (used as a test oracle).

5.5 Collision risk (Reich model)

$N = P_s(S) \cdot P_o(0) \cdot (\lambda_x / S_x) \cdot \{E_{\text{same}}[|\Delta v|/2\lambda_x + |v_o|/2\lambda_o + |v_s|/2\lambda_s] + E_{\text{opp}}[2|v|/2\lambda_x + |v_o|/2\lambda_o + |v_s|/2\lambda_s]\}$, with λ and S_x in NM and speeds in knots. Vertical: $P_s = P_z(S_z)$, $P_o = P_y(0)$, $v_o = \dot{y}$, $v_s = \dot{z}$. Lateral: $P_s = P_y(S_y)$, $P_o = P_z(0)$, $v_o = \dot{z}$, $v_s = \dot{y}(S_y)$. $P_y(S_y) \approx 2\lambda_y \cdot f_d(S_y)$, f_d the density of the difference of two independent deviations (Gaussian $\sigma\sqrt{2}$, or the convolution of two Laplace densities of scale b : $f_d(d) = (1 + |d|/b) \cdot e^{-|d|/b} / (4b)$). Minimum spacing is found by bisection on $\log N$.

5.6 Human error probability

$HEP = \min(1, GTT \times \Pi[(EPC_{\text{max}} - 1) \times APoA + 1])$. Library values are reference data with a version; changing them creates a new engine version.

5.7 Availability

Block $A = MTBF / (MTBF + MTTR)$; series ΠA_i ; parallel $1 - \Pi(1 - A_i)$; k-out-of-n by enumeration of up/down combinations. Block importance = $A(\text{system with block perfect}) - A(\text{system})$. Markov: build generator Q (off-diagonal rates, diagonal = $-\text{row sum}$), solve $\pi Q = 0$, $\Sigma \pi = 1$ by least squares; availability = $\Sigma \pi$ over up states; failure frequency = $\Sigma_{i \in \text{up}, j \in \text{down}} \pi_i q_{ij}$; mean down time = $U / \text{failure frequency}$; MTTF from $-Q_{uu} m = 1$.

5.8 Simulation statistics

Per condition: n , mean, sample standard deviation, 95% CI = $\text{mean} \pm t_{0.975, n-1} \cdot \text{sd} / \sqrt{n}$. Difference test: Welch's two-sample t-test (unequal variances), two-sided p-value. Criterion "threshold": solution $\text{mean} \leq \text{value}$ (lower is better) or $\geq \text{value}$; "no worse": difference $\leq \text{margin}$ in the worse direction.

5.9 Cooke's classical model and Delphi

Intrinsic range per item: $[\min - k(\max - \min), \max + k(\max - \min)]$ over all quantiles and the realisation, $k = 0.10$ (log scale for items so marked). Calibration $C_e = 1 - \chi^2_3(2N \cdot I(s, p))$ with $p = (0.05, 0.45, 0.45, 0.05)$ and s the observed inter-quantile proportions over N seeds; information = mean relative information of the expert's piecewise-uniform density against the uniform background; weight $\propto C \times I$ if $C \geq \alpha$, else 0; decision maker = weighted mixture of expert CDFs, quantiles by interpolation. Delphi: per round median, Q1, Q3 (linear interpolation), IQR, IQR/median, converging if IQR decreased.

5.10 Occurrence and security scores

ERC: index from the 4×4 matrix (Catastrophic 50/102/502/2500; Major 10/21/101/500; Minor 2/4/20/100; No accident 1/1/1/1 for barriers Effective/Limited/Minimal/Not effective). RAT: separation 0/1/3/7/10, closure 0/1/2/4/5, controllability items 0/1/2 each; score = sum. Security: $L \times \max(C, I, A)$.

5.11 Wildlife species risk

Per species over years Y : $n = \Sigma \text{strikes}$, $n_d = \Sigma \text{damaging}$; likelihood class from $n/|Y|$ with bounds >10 , ≥ 3 , ≥ 1 , ≥ 0.3 (classes 5-2, else 1); severity class from $100 \cdot n_d / n$ with bounds >20 , ≥ 10 , ≥ 5 , ≥ 1 when $n \geq 5$, otherwise from body mass with bounds ≥ 1.8 , ≥ 1.0 , ≥ 0.3 , ≥ 0.1 kg plus one class (maximum 5) for flocking species; score = $L \times S$; high if ≥ 12 , moderate if ≥ 6 , else low. Rate per 10,000 movements = $n / \Sigma \text{movements} \times 10^4$; trend = least-squares slope of the annual rate. Ranking by score, then by strikes.

5.12 Fuzzy AcciMap inference

Mamdani system per derived block. Evidence terms → crisp values SS 97.5, S 80.5, NE 50, O 19.5, SO 2.5; not provided → unknown. Input sets: trapezoids SO [-1, 0, 4, 5], O [5, 6, 33, 34], NE [34, 35, 65, 66], S [66, 67, 94, 95], SS [95, 96, 100, 101]. Hypothesis output triangles SO (0, 2.5, 5), O (5, 19.5, 34), NE (34, 50, 66), S (66, 80.5, 95), SS (95, 97.5, 100). Finding output triangles HU (0, 4.877, 10), IM (10, 21.5, 33), ML (33, 49.5, 66), PR (66, 78, 90), VP (90, 94.62, 99), AC (99, 99.5, 100). Rule degree = weight × min over antecedents of max membership over the listed terms (0 if the input is unknown); implication min; aggregation max; centroid on a 0.01 grid over [0, 100]; output term = set of highest membership. A block with no fired rule is not determined; its unknown inputs are clues with priority = share of its confirming rule's antecedents already satisfied. Hypothesis outputs propagate as crisp inputs.

5.13 Case-library Bayesian network and value of information

Naive Bayes structure per category: mechanism $C \rightarrow$ factor $F_i \in \{\text{present, absent}\}$. Approved cases give the mechanism (finding term $\geq$ Probable, highest) and the factors present (supported blocks). $P(C = c) = (n_c + 1)/(N + K)$; $P(F = \text{present} | c) = (n_{c,F} + 1)/(n_c + 2)$. Rating r of F : likelihood $\lambda(\text{present}) = \text{crisp}(r)/100$, $\lambda(\text{absent}) = 1 - \lambda(\text{present})$; NE and not provided add nothing. Posterior $P(c | e) \propto P(c) \prod_F [\lambda(p) \cdot P(p | c) + \lambda(a) \cdot P(a | c)]$. $\text{VOI}(F) = H(C | e) - \sum_f P(F = f | e) \cdot H(C | e, F = f)$ in bits, for unrated F , ranked descending. Export: rated factors get a child node observed “yes” with $P(\text{yes} | F) = \lambda(F)$.

5.14 Case search

Text normalised by replacing thesaurus phrases (longest first) with concept tokens, then lower-case word tokens without stop words, light suffix stemming. Weights $(1 + \ln \text{tf}) \times \text{idf}$ with $\text{idf} = \ln((1 + N)/(1 + \text{df})) + 1$ over the category's cases. Score = $0.6 \times \text{cosine} + 0.4 \times |\text{factors detected in the narrative} \cap \text{supported factors of the case}| / |\text{factors detected}|$, or the cosine alone if no factor is detected.

5.15 ATSB finding types and the 6×6 matrix

A test passes if its conclusion is “supported” and its probability expression, when given, has a lower bound $\geq 66\%$. Finding type: excluded if not analysed further; not established if existence fails; contributing if existence and influence pass and the influence target is the occurrence or a contributing factor (fixpoint over the chain); other if influence passes on a non-contributing factor, or importance passes; positive instead of contributing/other for positive actions and conditions; otherwise pending or not a safety factor. Compound warning if $p_{\text{existence}} \times p_{\text{influence}} < 50\%$ (lower bounds). ATSB 6×6: likelihood 1–6, consequence F–A with rank $F = 1 \dots A = 6$; $r = \text{likelihood} + \text{rank}$; critical $r \geq 10$, significant 8–9 (near critical) and 6–7, broadly acceptable $r \leq 5$. AirNav regions map intolerable → critical, tolerable (upper, lower) → significant, acceptable → broadly acceptable. Residual evaluation: no further action if residual is broadly acceptable, or significant and judged ALARP. Follow-up due = latest notification or communication date + 182 days.

6. Non-functional requirements

6.1 Performance and capacity (PER)

ID	Requirement	Pri.	Verif.
PER-01	Interactive pages and worksheets shall load within 2 s (95th percentile) for 100 concurrent users on the reference environment.	M	T
PER-02	Diagram editors shall remain responsive (under 200 ms per edit) for bowties of 60 barriers, fault trees of 500 events, FRAM models of 60 functions and BBNs of 60 nodes.	M	T
PER-03	Minimal cut sets for a 2 000-event tree shall be computed within 60 s; a fatigue simulation of one roster (28 days, 50 staff) within 10 s.	S	T
PER-04	The system shall hold at least 10 years of assessments (estimate 500 assessments and 20 000 hazard-log objects per year) without degradation.	M	A

6.2 Security (SEC)

ID	Requirement	Pri.	Verif.
SEC-01	Authentication shall use AirNav single sign-on (SAML 2.0 or OpenID Connect) with multi-factor authentication; local accounts only for break-glass administration.	M	T
SEC-02	Authorisation shall be role-based and scoped by organisational unit and confidentiality marking; the Administrator role shall not grant approval rights.	M	T
SEC-03	All traffic shall use TLS 1.2 or higher; data at rest (database, object store, backups) shall be encrypted with AES-256 or equivalent.	M	I
SEC-04	The application shall meet OWASP ASVS Level 2 and pass an independent penetration test before go-live and after major releases.	M	T
SEC-05	An append-only audit trail shall record creation, change (old/new value), approval, export and deletion of every object, with user and timestamp, retained for at least 10 years.	M	T

6.3 Privacy (PRV)

ID	Requirement	Pri.	Verif.
PRV-01	Personal data (participants, roster and sleep data, fatigue reports) shall be processed in line with [S4], with a documented purpose and retention period per data category.	M	I
PRV-02	Individual fatigue data shall be pseudonymised in all views except for the Fatigue analyst role, and exports of identifiable fatigue data shall require a recorded justification.	M	T
PRV-03	Participants in a study may be recorded by role instead of name where a just-culture policy requires it.	S	T

6.4 Availability, backup and recovery (AVL)

ID	Requirement	Pri.	Verif.
AVL-01	Availability shall be at least 99.5% during 06:00–22:00 WIB on working days, measured monthly, excluding planned maintenance announced 48 h ahead.	M	A
AVL-02	Backups shall allow recovery with a recovery point objective of 24 h and a recovery time objective of 8 h; restore shall be tested at least twice a year.	M	T

6.5 Usability, language and accessibility (USA)

ID	Requirement	Pri.	Verif.
USA-01	All UI text, help and generated report headings shall be available in Bahasa Indonesia and English, switchable per user.	M	I
USA-02	Each method module shall provide contextual help that links to the corresponding Manual chapter and shows the method's worked example.	S	D
USA-03	The UI shall conform to WCAG 2.2 level AA; colour shall never be the only carrier of meaning (risk regions also carry text labels).	S	T
USA-04	A new assessor with the Manual training shall complete a HAZID study and hazard-log entry in the tool without assistance within 30 minutes (usability test with at least five users).	S	T

6.6 Calculation assurance and maintainability (QA)

ID	Requirement	Pri.	Verif.
QA-01	Each calculation engine shall have a version number, a written specification (Section 5), unit tests with at least 90% branch coverage, and the reference test cases of Section 9 in its automated test suite.	M	I
QA-02	Engine verification shall be performed by a person or team independent of the engine's developer, and recorded before each release that changes an engine.	M	I
QA-03	Results stored with an assessment shall record the engine version; re-running with a newer engine shall create a new result, not overwrite the old one.	M	T
QA-04	Source code, build scripts and documentation shall be delivered to AirNav (or escrowed), with an OpenAPI description of all services.	M	I

7. External interfaces

Interface	Direction	Content	Mechanism	Pri.
Identity provider	In	Users, groups, MFA	SAML 2.0 / OIDC; SCIM for provisioning	M
Occurrence reporting / investigation system	In	Occurrence ID, date, unit, classification, barriers failed	REST API or scheduled file exchange	S
Rostering system	In	Planned and actual duty periods per staff member	REST API or CSV	M (for FAT)
CNS/ATM maintenance management	In	Equipment list, failure records, MTBF/MTTR	REST API or XLSX	S
Document management	Out	Approved Safety Assessment Reports	API upload with metadata	S
E-mail / notifications	Out	Assignments, reminders, approvals	SMTP; optional chat webhook	M
Public API	In/Out	All core objects	REST/JSON, OpenAPI 3, token-based	S

Import/export formats: XLSX/CSV (worksheets, hazard log), DOCX/PDF (reports), SVG/PNG (diagrams), Open-PSA MEF (fault trees), XMLBIF (BBN), FRAM Model Visualiser files (FRAM), documented JSON for bowties and full assessment archives.

8. Delivery phases

Phase	Content	Rationale
1 — Core (MVP)	COM, HZL, REP (basic), ADM, SEC, PRV; methods HAZID, HAZOP, JHA, Bowtie, FMEA (RPN)	Replaces the spreadsheets most used today and establishes the single hazard log and risk scheme.
2 — Quantitative	FHA, FTA engine, LOPA, FMECA criticality, frequency-to-likelihood mapping, dashboards	Supports system changes needing safety objectives and quantitative arguments.
3 — Systems and human performance	STPA, Fatigue model with roster integration, FRAM, BBN, occurrence integration	Adds methods for automation, complex socio-technical work and FRMS.
4 — Extended methods (Manual ed. 0.2)	CRM, ETA, HRA, ORC, GSN, CCA, RBD/Markov, SWIFT, HTA, SIM, SEJ, security risk (SRA), investigation (INV), wildlife strike risk (WLD), assessment templates (TPL), organisational function map (OFM), SPI register (SPI); links from investigations to bowtie barriers; GSN evidence links	Closes the method gaps identified in the edition 0.1 review: separation minima, human reliability, availability, occurrences, safety cases and security.

Each phase ends with acceptance testing against the requirements it delivers and the reference cases in Section 9 relevant to it.

9. Acceptance and reference test cases

The engines shall reproduce the worked examples of the Manual. Inputs are exactly those printed in the Manual; tolerances are relative unless stated.

Test	Input (Manual reference)	Expected result	Tolerance
TC-RISK-01	Severity B, likelihood 3 (Manual Ch. 3)	Index 3B; region Tolerable (upper); authority Accountable Executive	Exact
TC-RISK-02	Severity A, likelihood 3	Index 3A; Intolerable; approval blocked	Exact
TC-FMEA-01	Receiver degradation S=6, O=3, D=7 (Manual §10.3)	RPN 126; ranked first	Exact
TC-LOPA-01	f_{IE} 0.5/yr; P_{CM} 0.05; three IPLs PFD 0.1 (Manual §11.3)	$f_C = 2.5 \times 10^{-5}$ /yr; ratio to 1×10^{-5} target = 2.5	1e-9 abs.
TC-LOPA-02	As TC-LOPA-01 plus IPL PFD 0.1	2.5×10^{-6} /yr; target met	1e-9 abs.
TC-FTA-01	Tree of Manual Figure 7, probabilities as printed	Minimal cut sets {E4}, {E5,E6}, {E7,E8,E9}, {E1,E2,E3}; top = 1.2200×10^{-6}	0.1%
TC-FTA-02	Same tree	Fussell-Vesely: E4 0.820, E5 0.164, E7 0.0164; Birnbaum E5 0.0100	0.5%
TC-BBN-01	Network and CPTs of Manual §17.3	$P(E) = 0.00446$; $P(L) = 2.453 \times 10^{-4}$	1e-6 rel.
TC-BBN-02	Evidence L = yes	$P(F = \text{yes}) = 0.3901$; $P(W = \text{high}) = 0.5448$; $P(S = \text{no}) = 0.1091$	1e-4 abs.
TC-BBN-03	Evidence F = yes; then F = yes and W = high	$P(L) = 6.38 \times 10^{-4}$; 1.10×10^{-3}	1e-6 rel.
TC-FAT-01	Manual §15.4 no-nap roster; defaults of §5.1; $\Delta t = 5$ min; $S_0 = 13.0$ at 00:00 day 1 (asleep); sleeps 23:00–07:00 (days 1–2), 07:00–13:00 day 3	Night-duty minimum A = 5.18 at 05:55; max KSS 7.49; 39.6% of duty at $KSS \geq 7$	± 0.05 (A), ± 5 min, ± 1 pt
TC-FAT-02	As TC-FAT-01 plus nap 15:00–16:30 day 2	Minimum A = 6.30; max KSS 6.82; 0% at $KSS \geq 7$	As above
TC-FHA-01	Severity B, target region Tolerable (lower) (Manual §12.3)	Objective $\leq 1 \times 10^{-7}$ per operating hour	Exact
TC-CRM-01	Vertical parameters of Manual §18.4	Same 3.27×10^{-10} ; opposite 1.55×10^{-9} ; total 1.873×10^{-9} ; meets 2.5×10^{-9}	0.5%
TC-CRM-02	Lateral case of §18.4, double-exponential, b = 0.9 NM	Minimum spacing 11.68 NM; risk at 15 NM 1.58×10^{-10}	0.05 NM; 1%
TC-ETA-01	Level-bust tree of §19.3	S 2.25×10^{-2} ; F-S 2.0×10^{-3} ; F-F-S 4.5×10^{-4} ; F-F-F 5.0×10^{-5} per year	1e-9 rel.
TC-HRA-01	Tasks T1–T3 of §20.4	HEP 2.016×10^{-3} ; 5.75×10^{-2} ; 1.08×10^{-2}	1e-9 rel.
TC-ORC-01	Occurrences of §21.4	ERC 102 red, 50 amber, 1 green; RAT score 5 + 2 = 7	Exact
TC-RBD-01	Sector VHF RBD of §24.4	A = 0.9999517; downtime 0.4227 h/yr; VCCS most important	1e-6 abs.
TC-MKV-01	Standby Markov model, c = 0.99	U = 2.0396×10^{-6} ; mean down time 4.0 h	0.1%
TC-SEJ-01	Experts and items of §28.4, $\alpha = 0$	Weights A 0.814, B 0.186, C < 1e-4; DM median Q1 0.0815	0.005 abs.
TC-SEJ-02	Delphi rounds of §28.4	Medians 0.10; IQR 0.15, 0.04, 0.02; converging	1e-9
TC-SIM-01	Measures of §27.2	All criteria met; M1 means 3.6875 / 3.1625; Welch p < 0.001	1e-4
TC-SEC-01	Scenario SR-01 (L 3; C/I/A 1/5/3)	Score 15, Very high	Exact
TC-TPL-01	Create an assessment from template aim_acquisition	14 studies; HAZOP with 7 nodes and no likelihoods; FHA with 12 failure conditions; every GSN solution linked to a study of the new assessment	Exact
TC-AIM-01	FTA of Manual Appendix C (illustrative probabilities)	Top 2.857×10^{-5} per critical item; cut set {E2, E4, E5} 67%	0.1%

Test	Input (Manual reference)	Expected result	Tolerance
TC-TPL-02	Create an assessment from template org_change	11 studies (two STPA); function map 18 rows all “unknown”, 2 without a new owner; HAZID with organisational guidewords, no likelihoods	Exact
TC-ORG-01	DEMO-05 function map and Delphi	13 of 18 rows with gaps, 2 orphans; Delphi medians 0.125, 0.12, 0.12, IQR 0.125 → 0.015, converging	Exact; 1e-3
TC-PRJ-01	Project with one draft assessment, study, hazard and action; a second project with an accepted assessment	Archive hides it from the lists and blocks new assessments; restore reverses it. Delete: 403 for non-admin, 422 for a wrong code; removes 1 assessment, 1 study, 1 hazard, 1 action with an audit entry; 409 for the accepted project; a deleted demo is not re-seeded	Exact
TC-IES-01	Reference case of Ng et al. (2022) AcciMap 23, all eight evidence blocks Support	Hypothesis 80.5 Support; finding FE 94.54 Very probable, other mechanisms 4.959 Highly unlikely; with e2, e3 not provided: not determined, clues e2 and e3 with priority 0.5	0.01
TC-IES-02	Illustrative ATS library (9 cases), L-SIMILAR-CALLSIGN Strongly support	Prior M2 = 0.2; posterior equals the closed form of \$5.13; rated factor not among the clues; clues sorted by VOI	1e-4
TC-IES-03	Export of the network with three ratings	BBN engine posterior of C equals \$5.13 posterior	1e-6
TC-IES-04	Offline draft of a 7-sentence report	Draft case; mechanism M2; factors incl. similar callsign, congestion, hearback, callsign alert, LOS; all quotes verified; PDF upload gives the same drafter	Exact
TC-IES-05	AI draft with one invented quote and one unknown factor (mock service)	Invented quote flagged; unknown factor removed and kept as suggestion; invalid link dropped; rules generated	Exact
TC-ATB-01	ATSB 6×6 as configured	36 cells each in one level; 6C, 5B, 4A critical; 6F significant (lower); 3E, 1C broadly acceptable; every B and A cell at least significant; colours equal to the AirNav regions; AirNav 5A → critical, 3B → significant, 1E → broadly acceptable	Exact
TC-ATB-02	Eight factors incl. a chain, an excluded factor, a positive action and a test below the standard of proof	Contributing, other (influence on a non-contributing factor), other (importance), not established ×2, positive, excluded	Exact
TC-ATB-03	Factors with judgemental wording, no supporting item, wrong safety-issue type, duplicate sources, overdue follow-up	Each check raised; safety issue 2B significant; residual not ALARP → further action; ALARP → no further; follow-up 2026-08-02	Exact
TC-ATB-04	DEMO-06 ATSB analysis	7 contributing, 2 other, 2 positive; safety issues 3B (AirNav) and 2B (ATSB), both significant; no errors	Exact
TC-ATB-05	Expert system alignment	ATSB codes on catalogue factors; T lane accepted; “unsure” rating excluded from evidence and clues; ATSB probability labels	Exact
TC-ATB-06	DEMO-07 worked example (KNKT.24.10.22.04)	Factors for the KNKT contributing factors are contributing; vibration trigger other; 2 excluded; no errors or warnings; safety issues include F9, F10, F13-F16; all seven KNKT recommendations referenced	Exact
TC-ATB-07	Investigation report of DEMO-07; the same with the actions of F3 removed	Sections 1–6 with 3.1–3.5; corrective-action table columns as ATB-17, an elimination row and [Date]; every contributing factor in the table; each chain ends at the occurrence; F3 flagged row and check	Exact
TC-ATB-08	Report renderers and API	Factor map PNG; DOCX with section 5 and ≥ 4 images; PDF; JSON/PDF/DOCX from the endpoint; 422 for a non-ATSB study	Exact

Test	Input (Manual reference)	Expected result	Tolerance
TC-WLD-01	Species data of Manual §31.4 (2021–2025)	Bubulcus ibis ranked 1: 12.4/yr, 11.3% damaging, $5 \times 4 = 20$ high; Pteropus vampyrus severity 5 by mass surrogate; 1 high, 3 moderate, 2 low; 6.97 strikes per 10,000 movements	Exact; 1e-3

Verification of reference values. Values for TC-FAT-01/02 were computed with the simplified model (no brake, no ultradian term) and exact exponential updates per step, as specified in §5.1. The supplier shall confirm them independently before they are used for acceptance.

Appendix A. Traceability to the Manual

Manual chapter	Requirement groups
2 SRM process; 5 Workshops	COM-01 to COM-18
3 Risk classification scheme	COM-07 to COM-10, ADM-01, TC-RISK
4 Method selection	COM-03
6 Bowtie	BOW-01 to BOW-08
7 HAZID	HZD-01 to HZD-04
8 HAZOP	HZP-01 to HZP-04
9 JHA	JHA-01 to JHA-04
10 FMEA/FMECA	FME-01 to FME-05, TC-FMEA
11 LOPA	LOP-01 to LOP-05, TC-LOPA
12 FHA	FHA-01 to FHA-04, TC-FHA
13 STPA	STP-01 to STP-07
14 FTA	FTA-01 to FTA-08, §5.2, TC-FTA
15 Fatigue modelling	FAT-01 to FAT-08, §5.1, PRV, TC-FAT
16 FRAM	FRM-01 to FRM-05
17 BBN	BBN-01 to BBN-08, §5.4, TC-BBN
18 CRM	CRM-01 to CRM-05, §5.5, TC-CRM
19 ETA	ETA-01 to ETA-05, TC-ETA
20 HRA	HRA-01 to HRA-04, §5.6, TC-HRA
21 Occurrence risk classification	ORC-01 to ORC-05, §5.10, TC-ORC
22 GSN	GSN-01 to GSN-04
23 Common cause analysis	CCA-01 to CCA-03
24 RBD and Markov	RBD-01 to RBD-03, §5.7, TC-RBD, TC-MKV
25 SWIFT	SWF-01 to SWF-03
26 HTA	HTA-01 to HTA-03
27 Simulation	SIM-01 to SIM-04, §5.8, TC-SIM
28 Expert judgement	SEJ-01 to SEJ-04, §5.9, TC-SEJ
29 Security risk	SRA-01 to SRA-04, §5.10, TC-SEC
30 Investigation	INV-01 to INV-05
31 Wildlife strike risk	WLD-01 to WLD-08, §5.11, TC-WLD
Appendix C AIM template	TPL-01 to TPL-05, TC-TPL, TC-AIM
Appendix D organisational change	TPL-06, TPL-07, OFM-01 to OFM-03, SPI-01 to SPI-03, TC-TPL-02, TC-ORG-01
Appendix E investigation expert system	IES-01 to IES-15, §5.12-5.14, TC-IES-01 to TC-IES-05
Appendix F ATSB analysis	ATB-01 to ATB-19, §5.15, TC-ATB-01 to TC-ATB-08
32 Integration, hazard log, report	HZL-01 to HZL-05, REP-01 to REP-04
Technology stack (this SRS)	Appendix C

Appendix B. Glossary

Term	Meaning
Assessment	A documented safety assessment of one change or question, containing one or more method studies.
Method study	One application of one method (e.g. a HAZOP of the cut-over plan) within an assessment.
BDD	Binary decision diagram — data structure for exact Boolean function evaluation, used for fault-tree quantification.
MOCUS	Method for obtaining cut sets — classic top-down algorithm for minimal cut sets.
Open-PSA MEF	Open-PSA Model Exchange Format, an XML format for fault trees and event trees.
XMLBIF	XML Bayesian Interchange Format for Bayesian networks.
MoSCoW	Must / Should / Could (/ Won't) prioritisation.
Other terms	See the Manual, Section 1.3 and Appendix A.

Appendix C. Recommended technology stack

This appendix records the recommended technology stack for ARAP and how it compares with BowTieXP, the most widely used commercial bowtie tool, which is used here as a benchmark. It is a recommendation, not a requirement: a supplier may propose equivalents that meet Sections 4–7. The reference implementation published with this SRS (repository *ARAP---AirNav-Risk-Analysis-Platform*) follows this stack.

C.1 Benchmark: BowTieXP

BowTieXP (now part of the Wolters Kluwer Enablon portfolio as the “BowTieXP Suite”) began as a desktop editor and is now also offered as a browser-based enterprise platform with a central risk repository, barrier-performance monitoring, dashboards, multi-user collaboration, integrations, audit workflows and an incident module (IncidentXP). The capabilities ARAP must match or exceed are summarised below.

BowTieXP capability	How ARAP’s stack matches it
Bowtie editor with strong automatic layout (threats left, consequences right, barriers ordered on each path)	React Flow canvas with a purpose-built, deterministic bowtie layout routine; ELK.js for free-form diagrams (fault trees, STPA, BBN)
Rich barrier metadata and colour-coding (effectiveness, owner, type); barriers shared across paths	Custom React node components; a barrier is stored once and referenced on every path
Central repository, dashboards and barrier health	PostgreSQL as single source of truth; ECharts dashboards in the application; Apache Superset/Metabase on a reporting replica
Incident analysis linked to failed barriers	Occurrence-to-barrier links; barrier-failure counts shown on the bowtie
High-quality printing and export	Server-side SVG/PDF rendering with headless Chromium (Playwright); DOCX via templates
Scope: bowtie and incident analysis	ARAP extends to all 26 methods with one hazard log and one risk scheme

C.2 Recommended stack

Layer	Recommendation	Rationale / notes
Front end	React + TypeScript (Vite); Ant Design or MUI components; TanStack Query; Zustand; react-i18next (ID/EN)	Largest ecosystem and hiring pool; all serious diagram libraries support React
Diagrams	React Flow (@xyflow/react, MIT) + ELK.js; custom bowtie layout. Commercial alternatives: yFiles for HTML, JointJS+, GoJS	One engine for bowtie, FTA, STPA, FRAM and BBN. Prototype the bowtie editor with both free and commercial options before committing
Worksheets	AG Grid (Community; Enterprise if advanced features are needed)	Spreadsheet-like editing for HAZID, HAZOP, JHA, FMEA
Live workshops	WebSockets with record locking and presence; Yjs (CRDT) for true co-editing later	Keeps release 1 simple
Back end	Python 3.12 + FastAPI, modular monolith; Pydantic; SQLAlchemy 2 + Alembic; background jobs with Dramatiq or Celery + Redis	Calculation engines are natural in Python; OpenAPI generated automatically and typed client generated for the front end
Calculation engines	FTA: in-house BDD engine (SCRAM, GPL-3.0, as a test oracle only); BBN: exact variable elimination (in-house or pgmpy, MIT); fatigue, LOPA, FMECA, Monte Carlo: NumPy/SciPy	Versioned and independently verified (QA-01 to QA-03); reference cases of Section 9 in CI
Identity	Keycloak (OIDC/SAML, MFA, LDAP/AD federation), self-hosted	Meets SEC-01; can be hosted in Indonesia
Database	PostgreSQL 16+: relational core; JSONB for method models; recursive CTEs for trees; pg_trgm for duplicate detection and search; row-level security; trigger-based audit tables	No separate graph database needed: models have hundreds, not millions, of nodes
Files and cache	MinIO (S3-compatible) for attachments and reports; Redis for cache and job queue	Self-hostable

Layer	Recommendation	Rationale / notes
Reporting	docxtempl / python-docx for Word; Playwright (headless Chromium) for PDF	Diagrams print exactly as shown on screen
Analytics	Apache ECharts in the application; Apache Superset or Metabase (or Power BI if already standard) on a read-only reporting replica; JupyterHub for analysts	Operational dashboards in-app, organisation-wide BI outside the transactional database
Platform	Docker; Kubernetes (Docker Compose initially); OpenTelemetry, Prometheus/Grafana, Sentry; pytest, Vitest, Playwright; Semgrep, OWASP ZAP, dependency scanning	Meets AVL, QA and SEC groups

C.3 Alternative for .NET or Java organisations

If AirNav IT standardises on .NET or Java, keep the same front end and PostgreSQL, build the core services in ASP.NET Core or Spring Boot, and run the calculation engines as a separate Python service behind an internal API. This adds one service to operate but lets the existing team maintain the core.

C.4 Early decisions

1. **Free versus commercial diagram library.** React Flow is free and flexible but the bowtie layout, printing and undo must be built; yFiles or JointJS+ reduce that effort for a licence fee. The bowtie editor is the largest front-end item, so a two- to three-week prototype with both options is recommended before committing.
2. **Build versus hybrid.** A hybrid option is to license BowTieXP for bowtie work and build ARAP for the other eleven methods, importing BowTieXP XML exports into the ARAP hazard log. This is faster but means two licences and bowtie data held in two places.